# OMERO Course

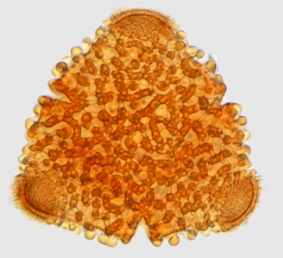
# Introduction to the Morphology of Pollen and Spores V. 1



**Carlos Jaramillo and the palynological team at STRI**

**Smithsonian Tropical Research Institute**

**Panamá**





**2026**

**A course sponsored by the**

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**This course is intended for novice practitioners of palynology.** This guide accompanies the images you will find in your Omero course.

Pollen and spores are tiny three-dimensional objects that require a light microscope to be observed. This presents a problem for beginners. It does not suffice to see a photograph of a pollen grain in a book, as the photograph depicts only a single focal plane of a three-dimensional structure. To observe an entire pollen grain from top to bottom, you would need to see between 20 and 200 focal planes depending on the size of the pollen grain. Therefore, observing the grains under a microscope is essential for learning to recognize their morphology.

The need for a microscope represents another obstacle to learning, as the optical quality of the microscope affects which features can be observed. For example, there are minute spines that an objective with a numerical aperture of 1.64 could resolve but they may not be visible with a numerical aperture of 0.6. Unfortunately, objectives with high numerical apertures are very expensive, and few institutions can afford them. Furthermore, to observe all the morphological traits present in pollen and spores, you will require a large reference collection that very few institutions in the world have. Lastly, the number of universities and research centers offering palynology courses have become increasingly rare.

All those problems represent a major setback for anyone who wants to learn about pollen and spore morphology. We believe that OMERO provides a practical solution to this problem. To build this course, we digitized hundreds of pollen and spore grains using the maximum possible numerical aperture in light microscopy (NA 1.6). Each pollen image is composed of a stack of tens to hundreds of focal planes separated by 0.3 microns. The entire stack is uploaded to OMERO, allowing you to visualize the grain plane by plane, as if you were looking through a microscope.

This course addresses all of these hurdles: being able to observe the grains in their three-dimensional nature, using images that closely resemble observations made with a high-end microscope, and having access to a large selection of grains in which each of the morphological characteristics of pollen and spore grains can be clearly observed.

Once you access OMERO using your credentials, you will see two main categories: Pollen and Spores.

The morphology of pollen and spores can be very complex with intricate patterns, leading to a vast array of descriptors that have been created to fit morphological features into categorical traits. Our discipline has been practiced for more than a century, and as the number of practitioners increased during the 20$^{th}$ century, so did the number of morphological terms. By the end of the 1980s, several morphological catalogues had been produced by different laboratories around the world containing literally thousands of morphological terms. It became evident that the terminology was getting out of hand, and over the past 20 years, considerable efforts have been made to unify these terms. This massive endeavor has resulted in a more standardized terminology.

Here, we follow the terminology of Punt et al 2007 (Punt et al., 2007), an excellent glossary of pollen and spore terms, where hundreds of terms are described, illustrated, and commented. Punt accepted and recommended 339 terms for use. A complementary glossary is the handbook by Hesse et al. 2009 (Hesse et al., 2009), which contains excellent SEM illustrations of hundreds of terms. If you plan to pursue a career in Palynology, I highly recommend consulting other excellent catalogues, including Erdtman 1952 (Erdtman, 1952), Kremp 1965 (Kremp, 1965), Iversen and Troels-Smith 1950 (Iversen & Troels-Smith, 1950), Faegri and Iversen 1989 (Faegri & Iversen, 1989), text books on Palynology including Traverse 2007 (Traverse, 2007) and Moore 1991 (Moore et al., 1991), two chapters of the AASP 1996 book, one for spores (Playford & Dettmann, 1996) and another for pollen (Jarzen & Nichols, 1996). There are also several electronic databases including PalDat < https://www.paldat.org/>, the Global Pollen Project < https://globalpollenproject.org/>, the taxonomy assistant in GeoGPT < https://geogpt.zero2x.org/tools/taxonomy-assistant>, PollenGeo < https://biogeodb.stri.si.edu/jaramillosdb/web/dashboard>, the Florida Tech Pollen Database < https://research.fit.edu/paleolab/pollen-database/>, and two useful non-electronic databases for fossil taxa, the Jansonius and Hill catalogue (Jansonius & Hills, 1976, 1981, 1985, 1987, 1990), and the Penn State Traverse catalogue (Traverse et al., 1985) among many others.

**This course is intended for novice practitioners of palynology.** It can be very intimidating and confusing to learn hundreds of morphological terms when you encounter pollen and spore grains for the first time. Therefore, we selected the most common terms (~130) used in the day-to-day work of a palynologist using a standard light microscope, working either with fossil or extant material.

We highly recommend keeping the Punt and Hesse catalogues on hand while taking this course. Read their definition of each of the terms in our course, and hand-draw the characteristics annotated as a ROI (Region of Interest) in the OMERO images. Hand-drawing is very important and cannot be replaced by simply visually inspecting the element; it is an effective tool for learning and remembering the terms.

Lastly, keep in mind that Nature does whatever it wishes. Morphological terminology attempts to capture a feature within a predefined box, but very often that feature does not neatly fit into a single category. Morphological categories are also subjective and can therefore lead to disagreements on which labels to use.

I hope the next generation of palynologists will embrace the challenge and use advances in digital imaging and quantitative analysis to develop better ways of describing pollen and spore morphology (Adaimé et al., 2025; Jaramillo et al., 2026; Mander et al., 2026; Stephenson, Shen, et al., 2026; Stephenson, Yang, et al., 2026) . Palynology is undergoing a fundamental transformation, and its terminology should evolve with it.

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**MORPHOLOGICAL TERMS IN THE OMERO COURSE**

Browse all the terms found in the course. Each green folder contains at least one example of the feature. For example, in Spores/A1_Aperture_Type/Trilete, you will find a spore with a trilete. Double click on the image to see it. You will find a slider on the left side that indicates the focal plane you are observing, and on the right side, you will see three tabs, Info, Settings, and ROIs. Click on the ROI tab, and you will see which ROIs are activated (ROI= Region of Interest). It will contain the feature that needs to be observed. In this case, you will see the trilete laesura. You can click on the 'show' button to active the ROI.

**Pollen**

The characteristics of a pollen grain can be divided into three main features: shape, aperture, and ornamentation.

**Shape** refers to the overall outline of the grain and includes several features

S1_Grain Association. It refers to how the grain is found. It could be isolated or still attached to its original tetrad.

- Monad
- Dyad
- Polyad
- Tetrad Linear
- Tetrad Decussate
- Tetrad Rhomboidal
- Tetrad Tetragonal
- Tetrad Tetrahedral

S2_Polar Pollen Shape. The outline of the grain from a polar view.

- Circular
- Elliptic
- Polygonal
- Rectangular
- Rhombic
- Triangular-Acute-Convex
- Triangular-Acute-Straight
- Triangular-Obtuse-Concave
- Triangular-Obtuse-Convex
- Triangular-Obtuse-Straight
- Trilobate

S3_Equatorial Pollen Shape. The outline of the grain from an equatorial view.

- Bisaccate
- Spherical
- Prolate

Oblate
Peroblate
Perprolate

S4_Polarity. Is the morphology of both extremes of the grain (e.g., polar areas) equal to each other?

Isopolar
Anisopolar

S5_Symmetry. How symmetric is the grain as seen from a polar view?

Radial
Bilateral
Asymmetric

**Aperture** refers to the number, structure, and shape of the apertures.

A1_Aperture Type. The number of apertures (colpus, pore) and how they combine with each other.

Inaperturate
Monoporate
Diporate
Triporate
Stephanoporate
Pantoporate
Monosulcate (Monocolpate)
Dicolpate
Tricolpate
Stephanocolpate
Pantocolpate
Zonasulculate
Tricolporate
Stephanocolporate
Pantocolporate
Syncolpate
Syncolporate
Trichotomosulcate
Dicolporate
Diploporate
Heterocolpate
Demicolpate

A2_Aperture Position. Are the apertures in the angular segments of the grains or in the planar segments as seen in polar view?

- Angulaperturate
- Planaperturate

A3_Aperture Variation. Are there variations in the wall structure near the apertures (e.g., thickenings, thinnings, openings)?

- Annulate
- Aspidate
- Atriate
- Costate
- Fastigiate
- Marginate
- Operculate
- Simple
- Vestibulate

A4_Pore Elongation. What is the shape of the pore?

- Circular
- Lolongate
- Lalongate
- Endocingulum

**Ornamentation** refers to the morphology of the wall, including elements that are above, inside, or below the wall.

O1_Exine Layers. How many layers have the wall?

- Atectate
- Intectate
- Semitectate
- Tectate

O2_Sculpture Type. How is the sculpture of the grain, both positive and negative features?

- Psilate
- Micropitted
- Foveolate
- Fossulate
- Foveoreticulate
- Reticulate
- Croton
- Scabrate
- Verrucate
- Baculate
- Gemmate

Clavate
Echinate
Rugulate
Striate
Striatoreticulate
Areolate
Fenestrate
Plicate
Elaterate

O3_Lumina Variation. Is there variation in the lumina across the grain surface?
Homobrochate
Heterobrochate

_____^^^_____

**Spore**
The characteristics of a spore grain can be divided into three main features including shape, aperture, and ornamentation.

**Shape** refers to the overall outline of the grain and includes several features.

S1_Lateral Shape. How is the shape from a lateral view?
Biconvex
Pyramidal
Plane-convex
Reniform

S2_Proximal/Distal Shape. How is the shape from a proximal or distal view?
Elliptic
Circular
Triangular-Obtuse-Convex
Triangular-Obtuse-Straight
Triangular-Obtuse-Concave
Triangular-Acute-Convex
Triangular-Acute-Straight
Triangular-Acute-Concave
Subtriangular-Concave-Expanded_Apices
Circular to Triangular-Obtuse-Convex

S3_Symmetry. How symmetric is the grain as seen from a proximal/distal view?

- Radial
- Bilateral

**Aperture** refers to the number, structure, and shape of the laesura.

A1_Aperture Type. What is the type of laesura?

- Alete
- Monolete
- Trilete

A2_Curvatura. How is the line delimiting the contact areas?

- Absent
- Imperfect
- Perfect

A3_Structural Modifications. How has the wall near the laesura been modified?

- Cingulate
- Zonate
- Congulizonate
- Valvate
- Auriculate
- Coronate
- Interradial-crassitude
- Kyrtomate
- Patinate
- Radial-appendages
- Polumbra
- Trifoliate
- Acrolamella

**Ornamentation** refers to the morphology of the wall, including elements that are above, inside, or below the wall.

O1_Exospore Layers. How many layers does the main wall of the spore have?

- 1-Layered
- 2-Layered

O2_Perispore Layers. In some cases, the perispore of the spore is preserved. Is it detached from the exosporte?

- Acavate
- Cavate

O2_Sculpture Type. How is the sculpture of the grain, both positive and negative features?

- Laevigate
- Micropitted
- Foveolate
- Fossulate
- Reticulate
- Foveoreticulate
- Scabrate
- Granular
- Verrucate
- Baculate
- Gemmate
- Clavate
- Echinate
- Rugulate
- Striate
- Cristate
- Cicatricosate
- Elaterate

# POLLEN

## A1. APERTURE TYPE:

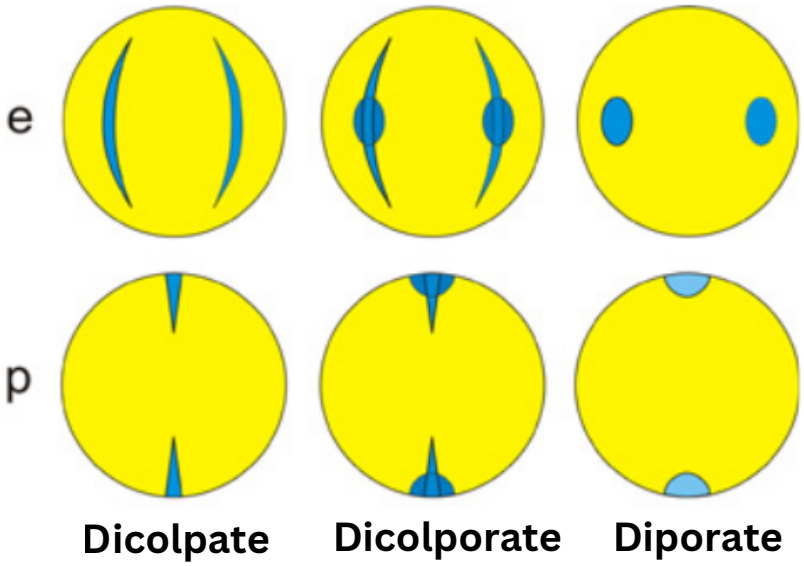
**Demicolpate**(adj.) (Erdtman, 1952)

An ectocolpus divided into two parts. Example: Amylotheca (Loranthaceae).

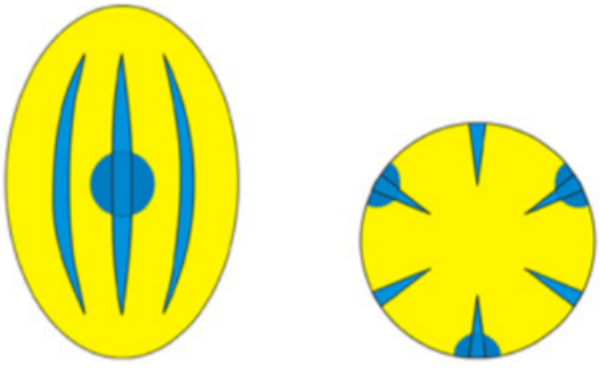


**Dicolpate, dicolporate, diporate** (adj.)
(Iversen and Troels-Smith, 1950)

Describing pollen grains with two ectocolpi, two compound apertures or two pores.



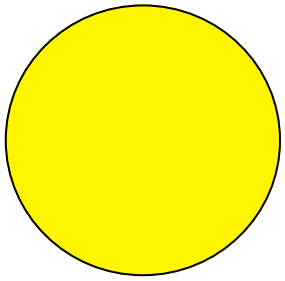

**Heterocolpate** (adj.) (Iversen and Troels-Smith, 1950)
Describing pollen grains with both simple and compound colpi present. Examples: *Lythrum, Peplis* (Lythraceae) and *Myosotis* (Boraginaceae).



**Inaperturate** (adj.) (Iversen and Troels-Smith, 1950)

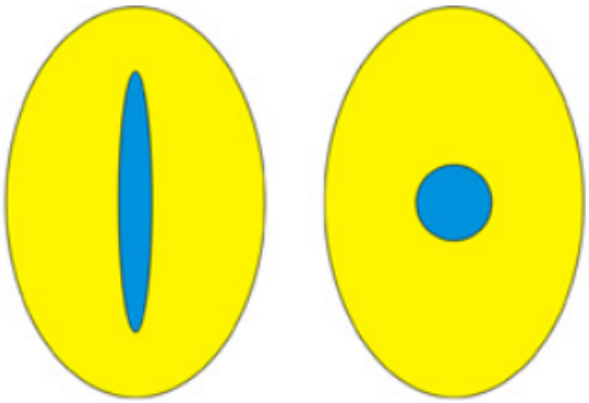
Describing a pollen grain without apertures. Example: *Populus* (Salicaceae), *Taxus* (Taxaceae).

**Comment:** The term should be used where apertures are completely absent. If ectoapertures are absent, but endoapertures are present, the pollen is cryptoaperturate.



**Monoaperturate** (adj.)

Describing a pollen grain with a single aperture.




**Pantoaperturate** (adj.) (Erdtman and Vishnu-Mittre, 1956)

Describing a pollen grain with apertures spread over the surface sometimes forming a regular pattern.

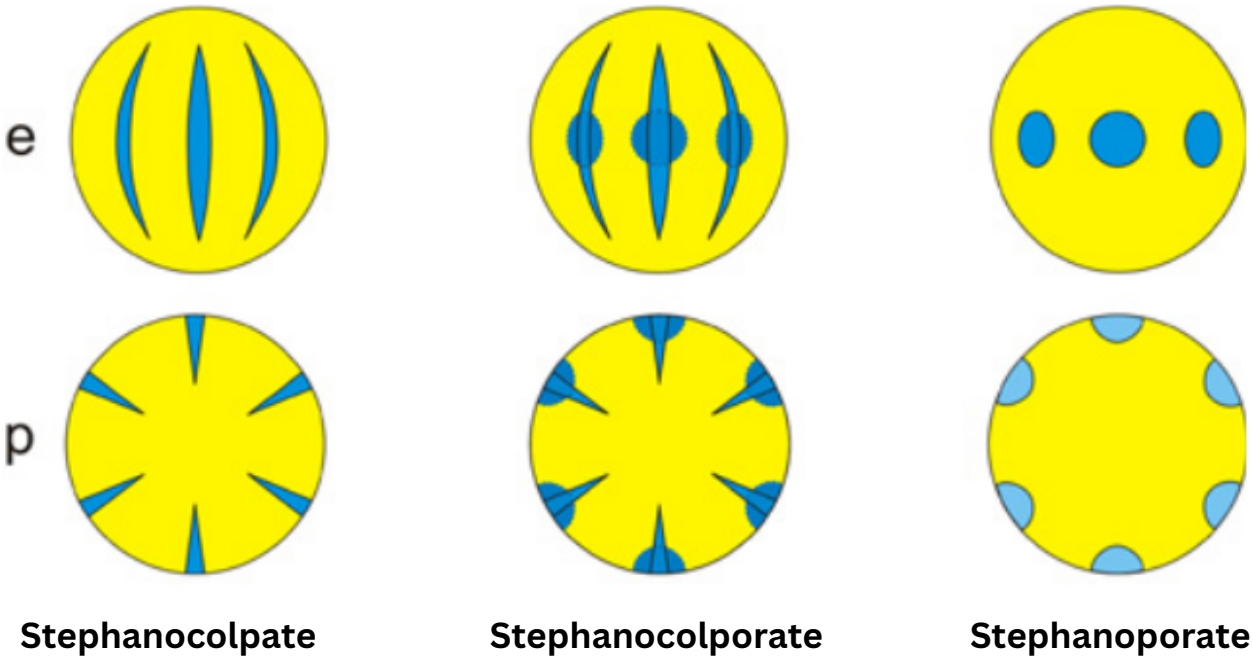
**Comment:** Such pollen grains may be, for example, pantocolpate, pantocolporate or pantoporate.



**Stephanoaperturate or Zonoaperturate**
(adj.) (Erdtman and Straka, 1961)

Describing a pollen grain with apertures situated only at the equator.

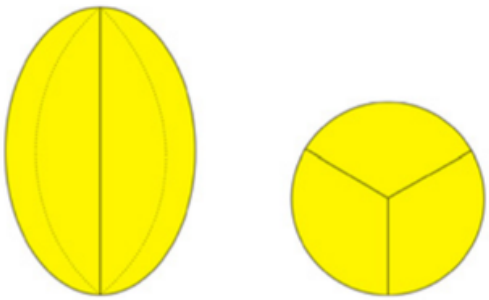
**Comment:** Zono- synonym of stephano-




**Stephanocolpate** **Stephanocolporate** **Stephanoporate**

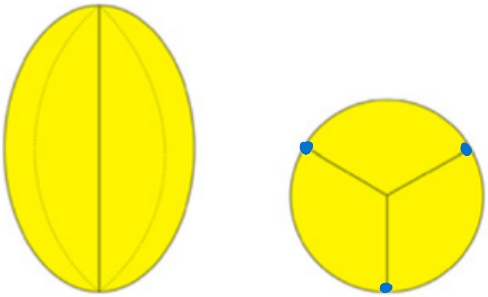
**Syncolpate** (adj.)

Pollen grain with anastomosing colpi.



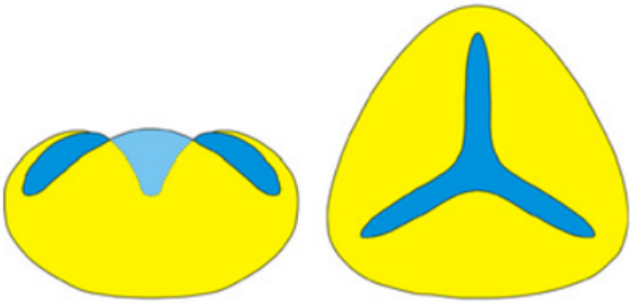
**Syncolporate** (adj.)

Pollen grain with anastomosing colpori.



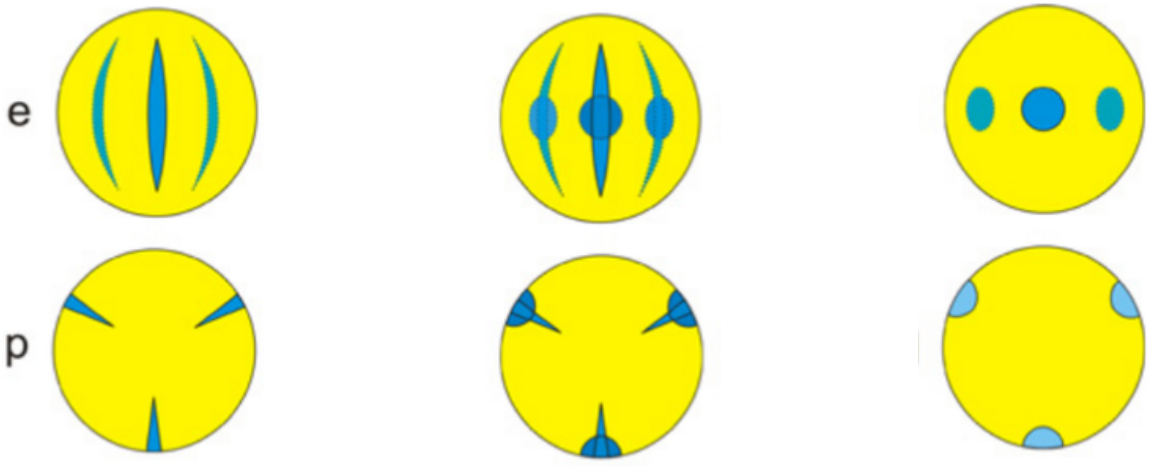
**Trichotomosulcate** (adj.) (Erdtman, 1952)

Describing a pollen grain with a three-armed sulcus. Example: *Elaeis guineensis* (Palmae).

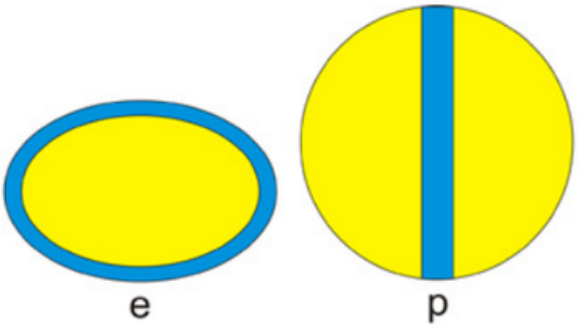


**Tricolpate, tricolporate, triporate** (adj.)
(Iversen and Troels-Smith, 1950)

Describing pollen grains with three ectocolpi, three compound apertures or three pores.



**Zonasulculate** (adj.) (Erdtman, 1952)
A meridional ring-like sulcus perpendicular to the equator. Example: *Laurelia novaezelandiae* (Monimiaceae).

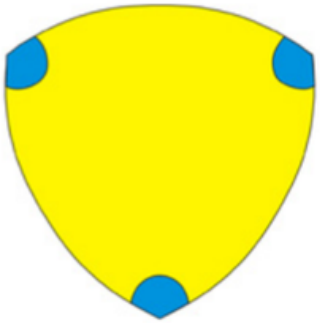



## A2. APERTURE POSITION:

**Angulaperturate** (adj.) (Erdtman, 1952)

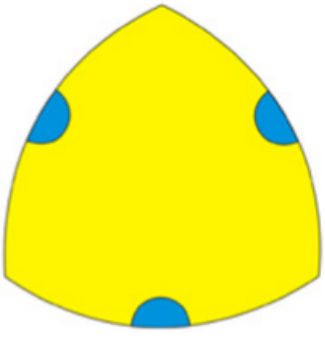
Describing an equatorially aperturate pollen grain with the apertures situated at the angles of the outline in polar view. Example: *Corylus* (Betulaceae).

> **Comment:** The term is useful for describing the position of apertures, but should be avoided as a description of equatorial outline.



**Planaperturate** (adj.) (Erdtman, 1952)
Describing a pollen grain with an angular outline, in which the apertures are situated in the middle of the sides when seen in polar view, rather than at the angles. Example: *Tilia* (Tiliaceae).

> **Comment:** The term is useful for describing the position of apertures, but should be avoided as a description of equatorial outline.

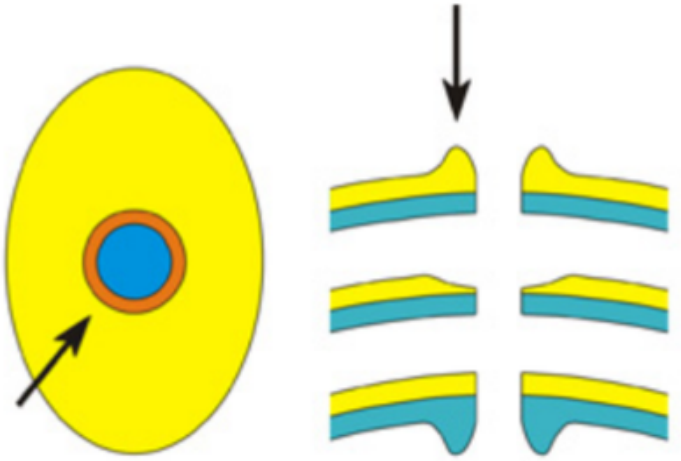


## A3. APERTURE VARIATION:

**Annulate** (adj.)

An area of the exine surrounding a pore that is sharply differentiated from the remainder of the exine, either in ornamentation or thickness (Jackson, 1928).

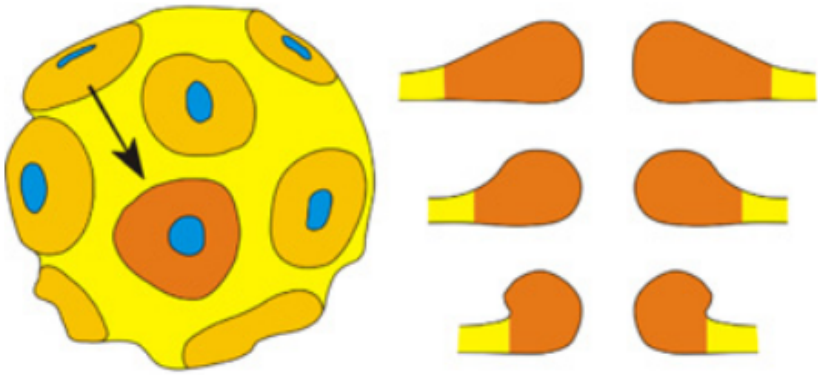
> **Comment:** Although the correct Latin spelling is anulus, the form used throughout the botanical literature is annulus.



**Aspidate** (adj.) (Wodehouse, 1935)

A prominently protruding thickening of the exine around a pore. Examples: *Betula* (Betulaceae), *Dorstenia* (Moraceae).

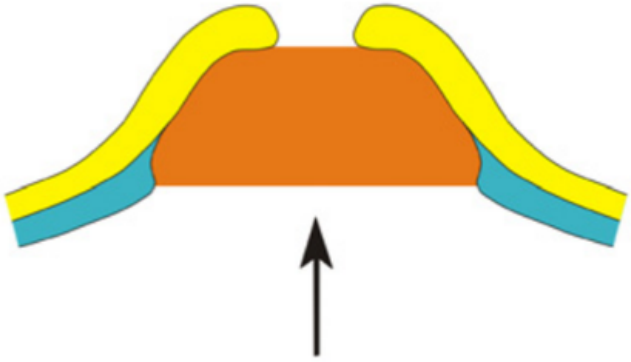
> **Comment:** An aspis is a special form of annulus, represented by a thickening rather than a thinning.



**Atriate** (adj.) (Thomson and Pflug, 1953)

A space within the aperture of a compound pore that has a much larger endopore than the ectopore, so that the pore

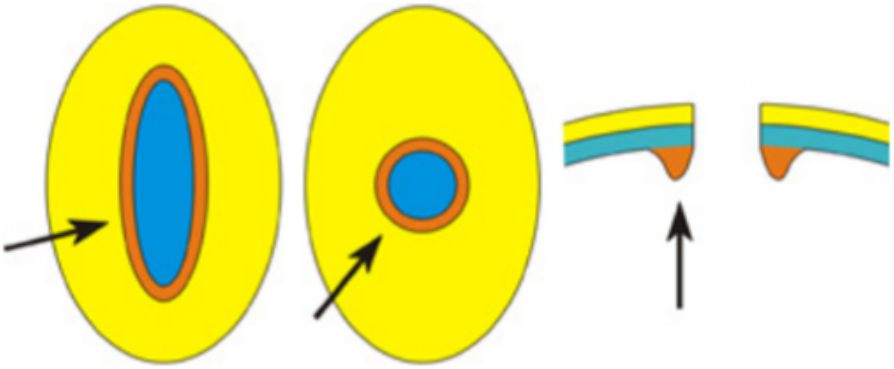
canal widens towards the interior of the grain. Example: *Myrica* (Myricaceae).



**Costate** (adj.) (Iversen and Troels-Smith, 1950)

A thickening of the nexine/endexine bordering an endoaperture, or following the outline of an ectoaperture.

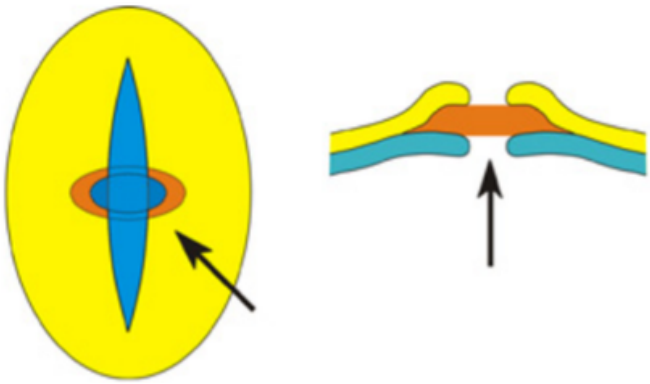
> **Comment:** The following forms are used: costa ectocolpi, costa ectopori, costa endocinguli, costa endocolpi, costa endopori.



**Fastigiate** (adj.) (Reitsma, 1966)

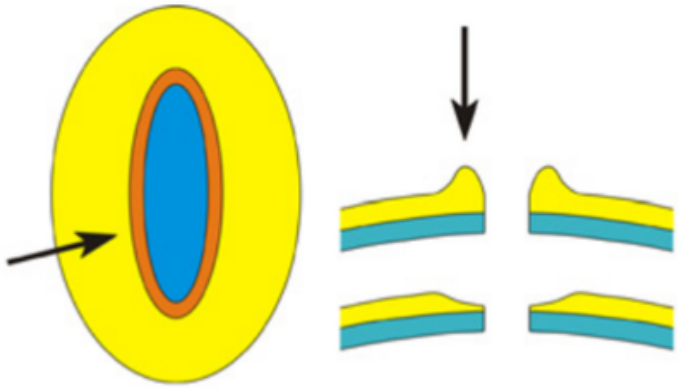
Cavity in a colporate grain, appearing as a separation of the inner part of the exine from the domed sexine in the region of the endoaperture. Example: *Hypericum* (Hypericaceae).

> **Comment:** The term describes a feature morphologically similar to the vestibulum; the latter is restricted to porate pollen grains.



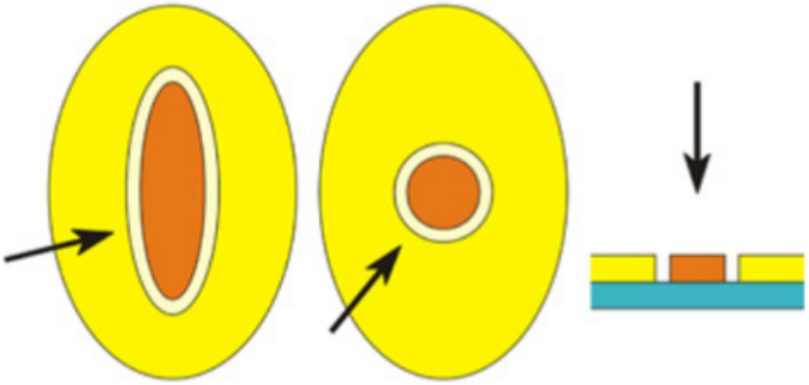
**Marginate** (adj.) (Iversen and Troels-Smith, 1950)

An area of exine around an ectocolpus that is differentiated from the remainder of the sexine, either in ornamentation or by difference in thickness.



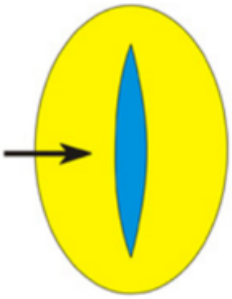
**Operculum** (adj.) (Wodehouse, 1935)

A distinctly delimited sexineectexine structure which covers part of an ectoaperture and which is completely isolated from the rest of the sexine.



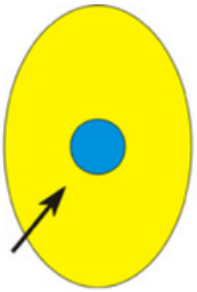
**Simple**

**Colpus (adj. colpate) (Erdtman, 1943):** An elongated, aperture with a length/breadth ratio greater than 2.



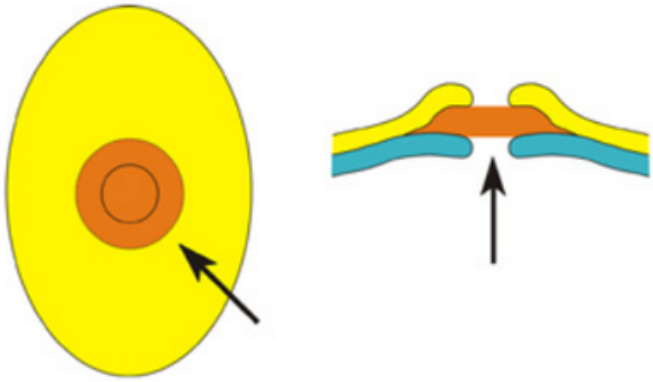
**Pore (adj. porate) (Jackson, 1928; Wodehouse, 1935)**

A general term, applied in palynology to a circular or elliptic aperture with a length/breadth ratio less than 2.



**Vestibulate** (adj.) (Potonié, 1934)

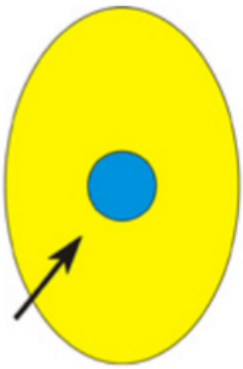
A separation between layers of the exine forming a cavity between the inner and outer pores. For example: *Betula* (Betulaceae).



## A4. PORE ELONGATION:

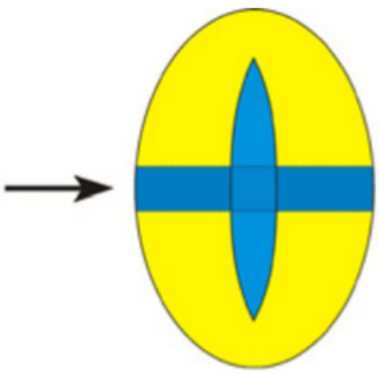
**Circular** (Wodehouse, 1935)

A general term, applied in palynology to a circular aperture with a length/breadth ratio less than 2.



**Endocingulum** (adj. endocingulate) (Reitsma, 1966)

A general term, applied in palynology to a circular or elliptic aperture with a length/breadth ratio less than 2.



**Lalongate** (adj.) (Erdtman, 1952)

Describing the shape of a transversely elongated endoaperture. Example: *Filipendula* (Rosaceae).



**Lolongate** (adj.) (Erdtman, 1952)

Describing the shape of a longitudinally elongated endoaperture. Example: *Rumex* spp. (Polygonaceae).



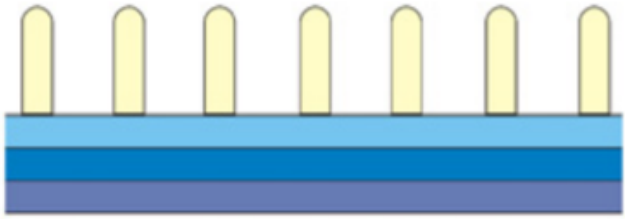
## O1. EXINE LAYERS:

**Atectate** (adj.) (Walker and Doyle, 1975)
Describing pollen grains that have an exine with little or no internal structure. Example: *Degeneria* (Degeneriaceae).

> **Comment:** The term was originally intended to describe walls of primitive angiosperms lacking columellae. Related terms include intectate, which is used for pollen grains with columellae but without a tectum, and etectate which is applied to pollen grains believed to have lost their tectum during phylogeny.

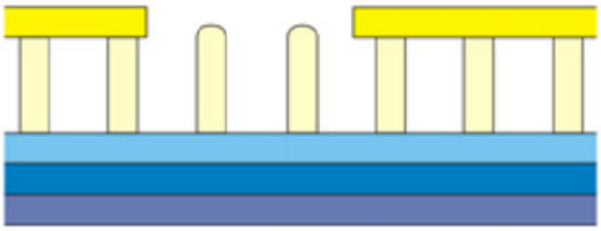


**Intectate** (adj.) (Iversen and Troels-Smith, 1950)
Describing pollen grains without a tectum, but with sculpturing. Examples: *Viscum* (Loranthaceae), *Ilex* (Araliaceae).



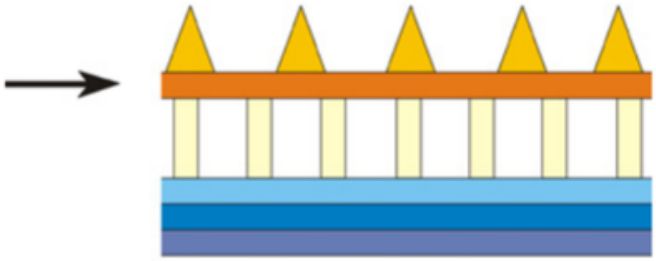
**Semitectate** (adj.) (Fægri and Iversen, 1964)

A partially discontinuous tectum in which the tectal perforations are equal to or wider than the muri and usually larger than 1 µm in diameter.



**Tectate** (adj.) (Iversen and Troels-Smith, 1950)

The layer of sexine, which forms a roof over the columellae, granules or other infratectal elements.

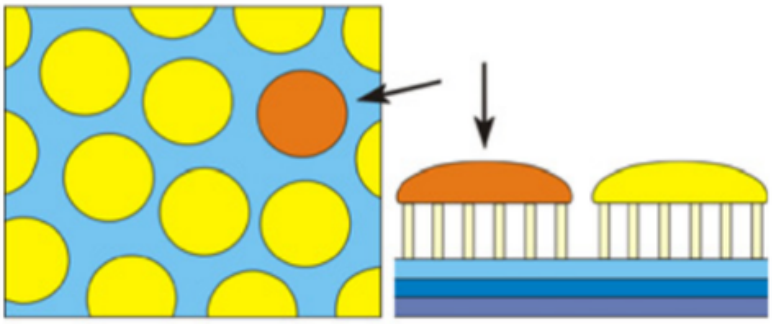


## O2. SCULPTURE TYPE:

**Areolate** (adj.) (Erdtman, 1947)

A feature of ornamentation in which the sexine/ectexine is composed of circular or polygonal areas separated by grooves

which form a negative reticulum. Examples: *Apama* (Aristolochiaceae), *Phyllanthus* spp. (Euphorbiaceae).



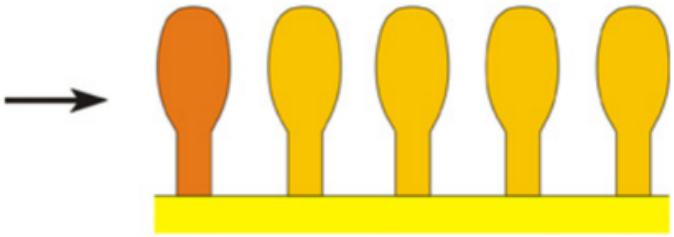
**Baculate** (adj.) (Potonié, 1934)

A cylindrical, free standing exine element more than 1 µm in length and less than this in diameter. Example: *Raistrickia saetosa*.



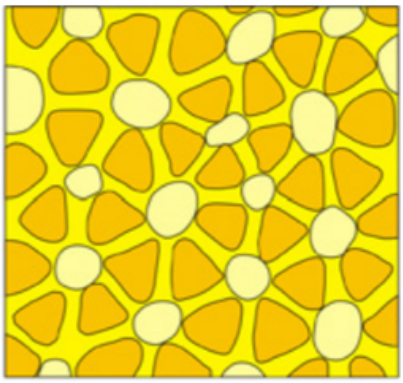
**Clavate** (adj.) (Iversen and Troels-Smith, 1950)

A club-shaped element of the sexine/ectexine that is higher than 1 µm, with diameter smaller than height and thicker at the apex than the base.



**Croton pattern** (Erdtman, 1952)

A characteristic type of ornamentation comprising rings of five or six (sometimes more) raised, often triangular, sexine elements arranged around a circular area, usually formed by capitate columellae (pila). Example: *Croton, Jatropha* (Euphorbiaceae), *Callitriche antartica* (Callitrichaceae), *Pimelea arenaria* (Thymelaceae).



**Echinate** (adj.) (Wodehouse, 1928)

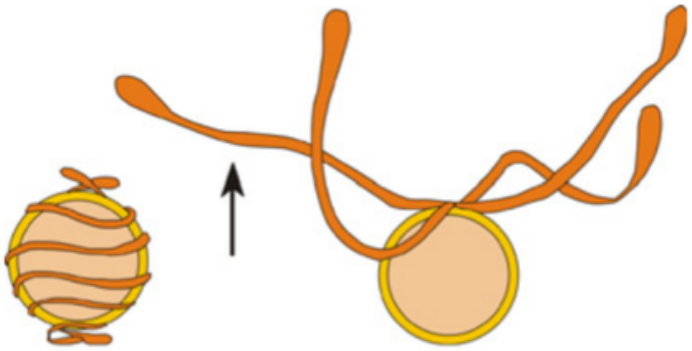
Describing pollen and spores with an ornamentation comprising spines longer than 1 µm

> **Comment:** (Erdtman (1952) recommended the term spinose, but in his usage spines were defined as longer than 3 µm and smaller features as spinules (adj. spinulose).



**Elater**

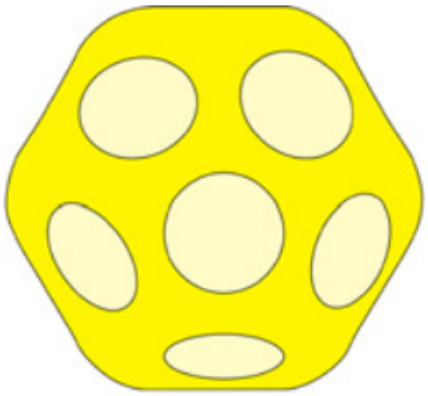
A spirally twisted, filamentous band attached to certain spores and some fossil pollen grains (Jackson, 1928). Example: *Equisetum* (Equisetaceae).



**Fenestrate** (adj.) (Iversen and Troels-Smith, 1950)

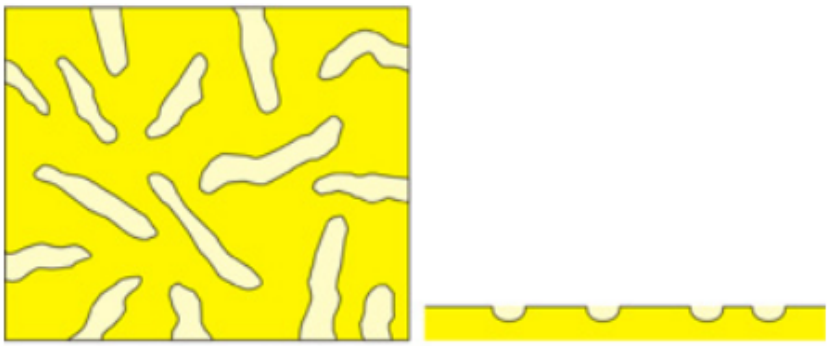
Describing a class of pollen grains characterized by large, window-like spaces lacking a tectum.

**Comment:** The term is accepted as a category in the classification system of Iversen and Troels-Smith (1950) and was especially established to include lophate pollen grains of the Compositae. Although useful for defining a class of pollen grains the term is not recommended in descriptions. Many such grains can be described as lophate.



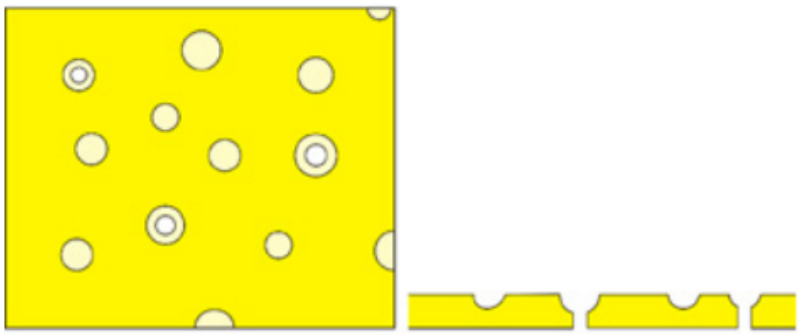
**Fossulate** (adj.) (Fægri and Iversen, 1950)

A feature of ornamentation consisting of an elongated, irregular groove in the surface.



**Foveolate** (adj.) (Erdtman, 1952)

A feature of ornamentation consisting of more or less rounded depressions or lumina more than 1 µm in diameter. The distance between foveolae is greater than their breadth.



**Foveoreticulate**
Describes a pollen surface that has pits large enough and close enough together to form a network-like pattern.

**Gemmate** (adj.) (Iversen and Troels-Smith, 1950)

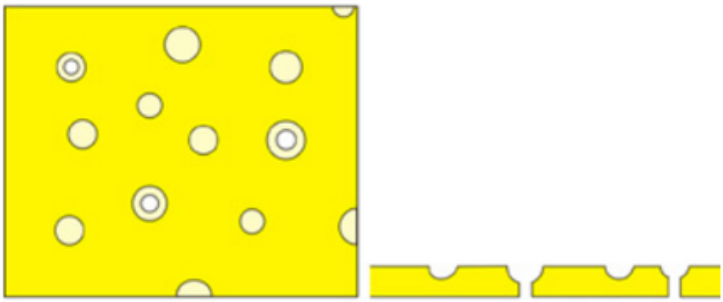
A sexine element which is constricted at its base, higher than 1 µm, and that has approximately the same width as its height.



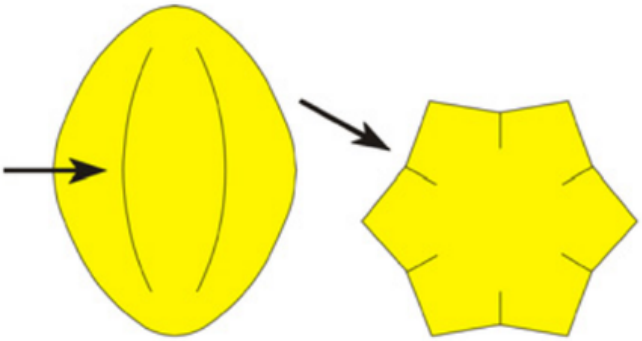
**Micropitted** (adj.)

A general term for small depressions (Jackson, 1928).

**Comment:** In palynology, the prefix micro- generally used to denote features less than 1 µm.

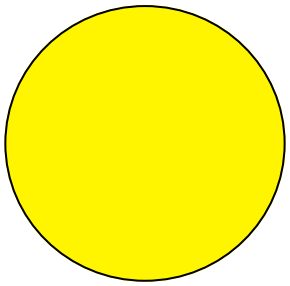


**Plicate** (adj.) (Thomson and Pflug, 1953)
A general term for a fold, applied in palynology to ridge-like folds of the exine in *Ephedra* (Ephedraceae) and *Lusatipollis*.



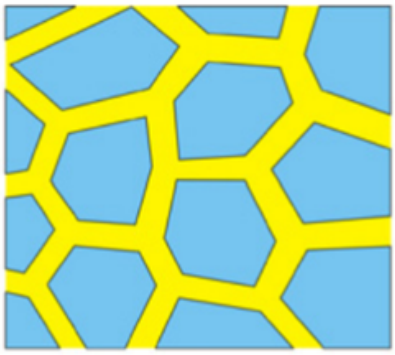
**Psilate** (adj.) (Wodehouse, 1928)

Describing a pollen or spore with a smooth surface.



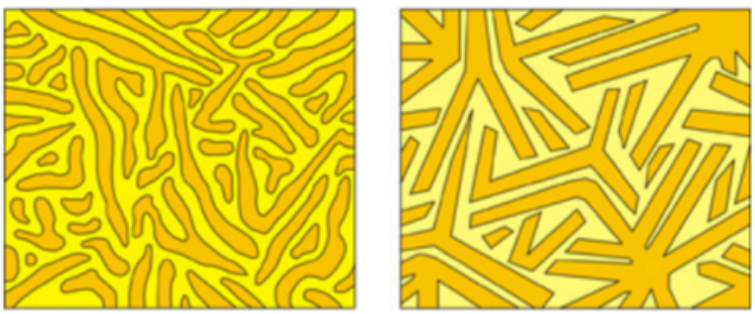
**Reticulate** (adj.) (Praglowski and Punt, 1973)

A network-like pattern consisting of lumina or other spaces wider than 1 µm bordered by elements narrower than the lumina.



**Rugulate** (adj.) (Iversen and Troels-Smith, 1950)

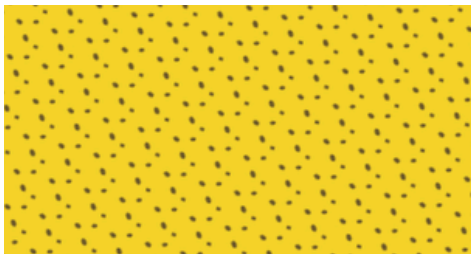
Describing a type of ornamentation consisting of elongated sexine elements more than 1 µm long, arranged in an irregular pattern that is intermediate between striate and reticulate. Examples: *Sedum* (Crassulaceae), *Ulmus* (Ulmaceae), *Camptotriletes corrugatus*.



**Scabrate** (adj.) (Iversen and Troels-Smith, 1950)

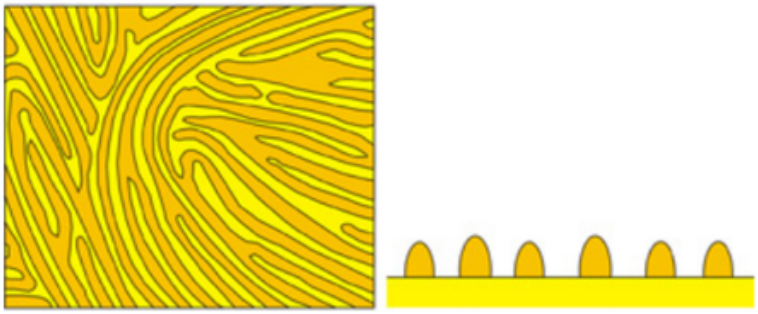
Describing elements of ornamentation, of any shape, smaller than 1 µm in all directions. Examples: *Quercus* (Fagaceae), *Artemisia* (Compositae).

**Comment:** Ornamentation elements larger than 1 µm are described according to their shape, for example, baculum, clava, gemma, verruca.

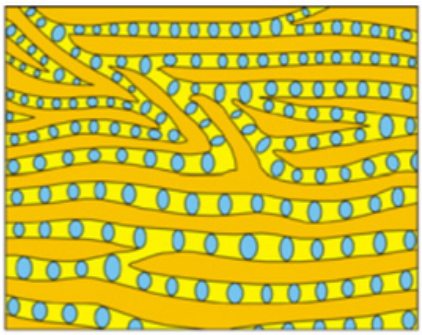


**Striate** (adj.) (Iversen and Troels-Smith, 1950)
A general descriptive term applied in palynology to elongated, generally parallel elements separated by grooves.

**Comment:** The positive elements of striate ornamentation may also be referred to as muri.



**Striato-reticulate** (adj.) (Erdtman, 1952)
Describing a pattern in which parallel or subparallel muri are cross-linked to form a reticulum in the grooves. The connections between the muri lie on a single level or different levels. Example: *Gentiana pneumonathe* (Gentianaceae).



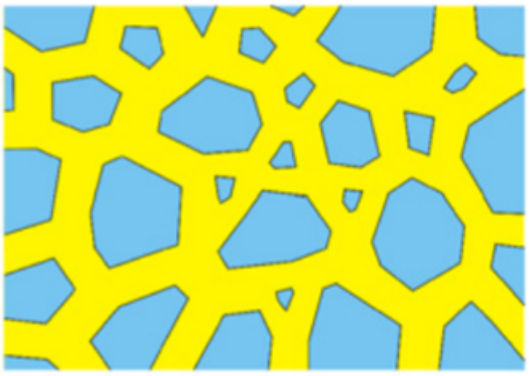
**Verrucate**(adj.) (Iversen and Troels-Smith, 1950)
A wart-like sexine element, more than 1 µm wide, that is broader than it is high and is not constricted at the base. Example: *Plantago* (Plantaginaceae).



## O3. LUMINA VARIATION:

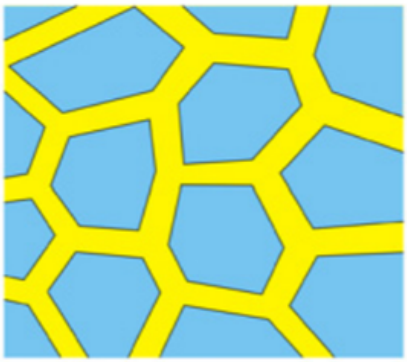
**Heterobrochate** (adj.) (Erdtman, 1952)

Describing a reticulum with brochi of different sizes. Examples: *Adoxa moschatelina* (Adoxaceae), *Forsythia europaea* (Oleaceae).



**Homobrochate** (adj.) (Erdtman, 1952)

Describing a reticulum with brochi of the same sizes. Example: *Armeria maritima* (Plumbaginaceae).

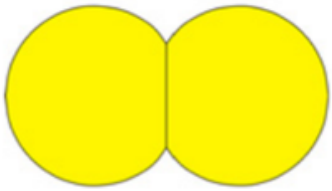


## S1. GRAIN ASSOCIATION:

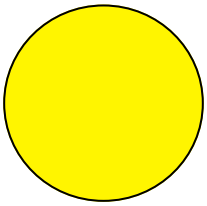
**Dyad**

General term for two microspores (pollen grains or spores) united as a dispersal unit.

**Comment:** A developmental dyad stage occurs prior to tetrad formation in plants with successive meiosis.

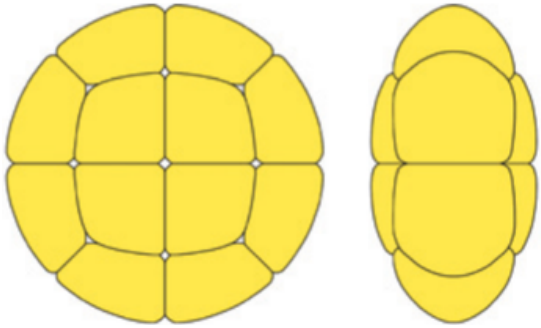


**Monad** (Selling, 1947)
A pollen grain or spore dispersed as an individual unit, rather than in association with others, such as in a dyad, tetrad or polyad.



**Polyad** (Iversen and Troels-Smith, 1950)

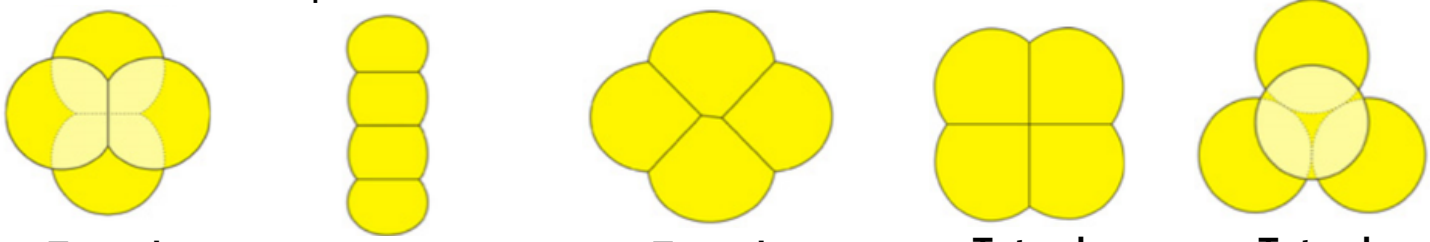
A dispersal unit comprising more than four pollen grains. Example: *Acacia* (Mimosaceae).



**Tetrad**

A general term for a group of four united pollen grains or spores, either as a dispersal unit or as a developmental stage.

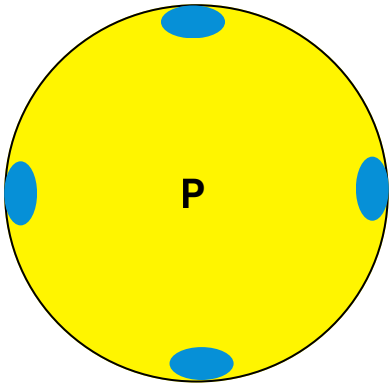
**Comment:** Tetrads may be uniplanar, with all members lying in the same plane or multiplanar, with members in more than one plane



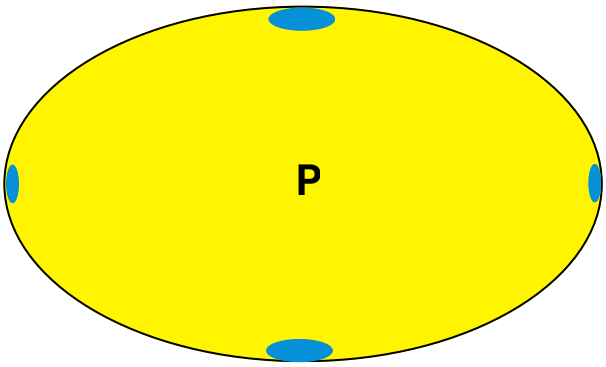
**Tetrad decussate** **Tetrad linear** **Tetrad rhomboidal** **Tetrad tetragonal** **Tetrad tetrahedral**

## S2. POLAR POLLEN SHAPE:

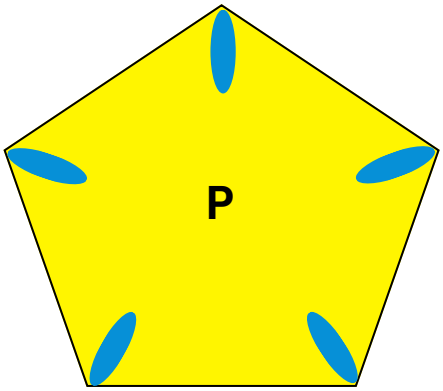
**Circular**
A pollen grain with an elliptical outline.




**Elliptic**
Oval outline, longer than it is wide.

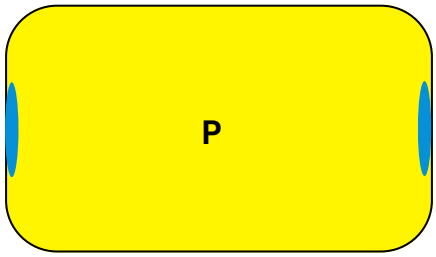



**Polygonal**
A pollen grain with a many-sided, angular outline.



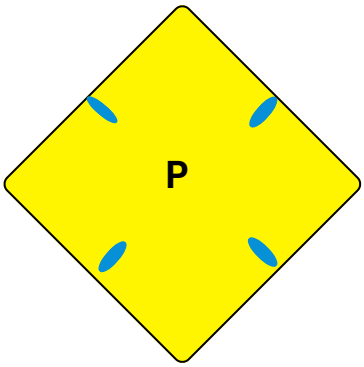
**Rectangular**

A pollen grain with an elongated, four-sided outline and roughly parallel sides.



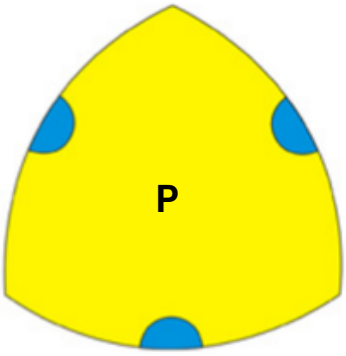

**Rhombic**

A pollen grain with a diamond-shaped outline.



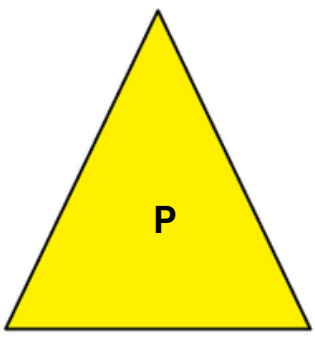

**Triangular-Acute-Convex**

Sharp angles, sides bulged outwards.



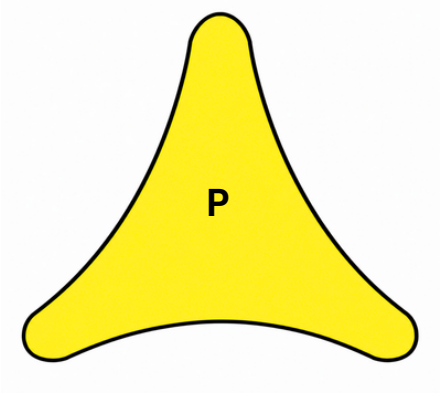

**Triangular-Acute-Straight**

Sharp angles, sides bulged outwards.



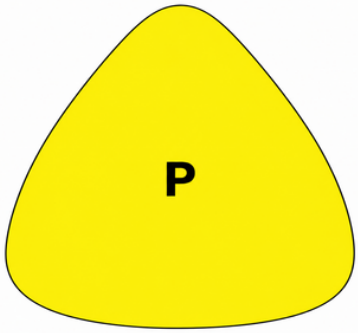
**Triangular-Obtuse-Concave**

Blunt rounded angles, sides curved inwards.



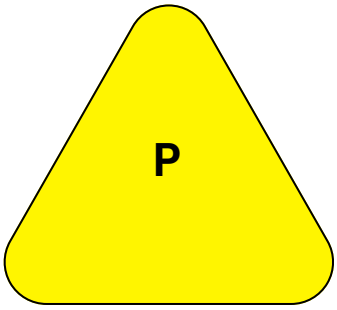

**Triangular-Obtuse-Convex**

Blunt rounded angles, sides bulged outwards.



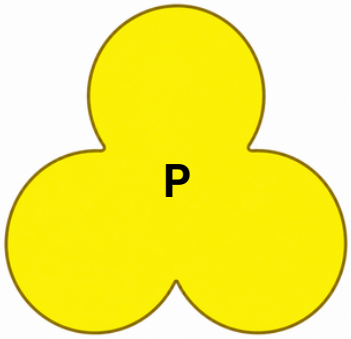

**Triangular-Obtuse-Straight**

Blunt rounded angles, flat sides.




**Trilobate**

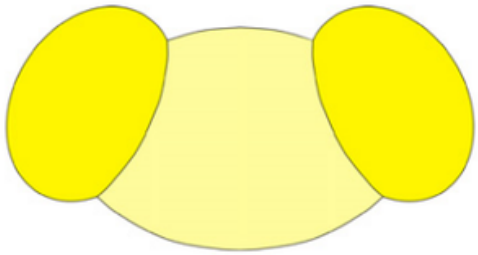
Blunt rounded angles, flat sides.



## S3. EQUATORIAL POLLEN SHAPE

**Bisaccate** (adj.) (Potonié and Kremp, 1954)

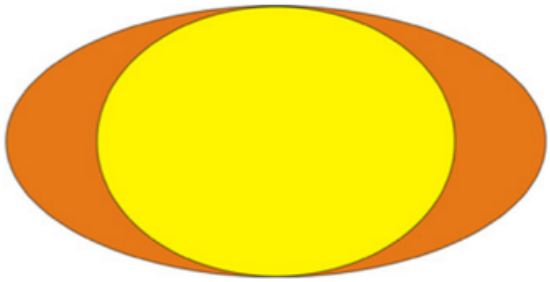
Describing pollen with two sacci. Example: *Pinus* (Pinaceae).



**Oblate** (adj.) (Erdtman, 1943)

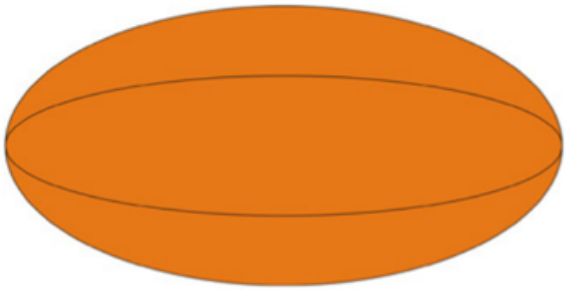
Describing the shape of a pollen grain in which the polar axis is shorter than the equatorial diameter.

> **Comment:** In this system oblate is defined as a ratio between the polar axis and the equatorial diameter of 0.50–0.75.



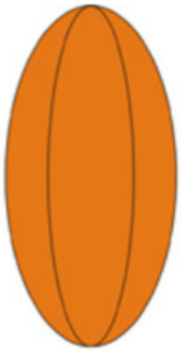
**Peroblate** (adj.) (Erdtman, 1943)

Describing the shape of a pollen grain in which the ratio between the polar axis and the equatorial diameter is less than 0.50.



**Perprolate** (adj.) (Erdtman, 1943)

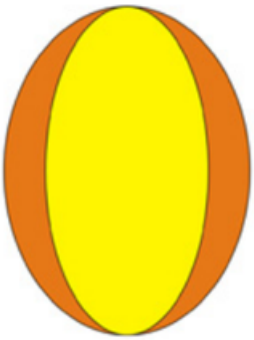
Describing the shape of a pollen grain in which the ratio between the polar axis and the equatorial diameter is more than 2.



**Prolate** (Erdtman, 1943)

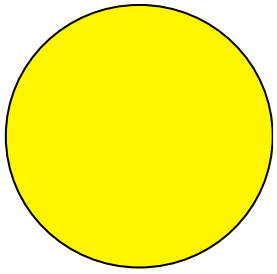
Describing the shape of a pollen grain in which the polar axis is larger than the equatorial diameter.

> **Comment:** In this system prolate is defined as a ratio between the polar axis and the equatorial diameter of 1.33 - 2.00.



**Spherical**

Describing the shape of a pollen grain in which the polar axis is equal to the equatorial diameter.

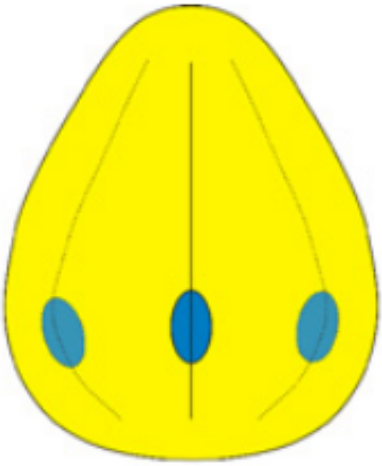


## S4. POLARITY

**Anisopolar** (adj.) (Erdtman, 1947)

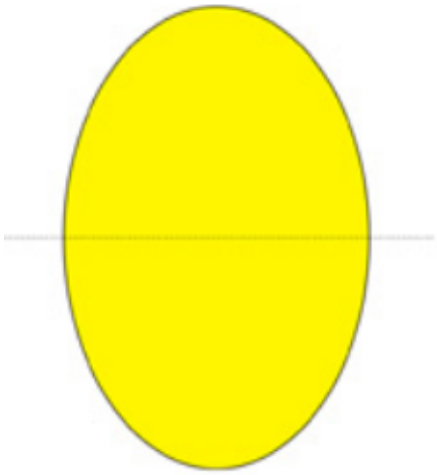
Describing pollen or spores in which the distal and proximal faces of the exine are different, either in shape, ornamentation or apertural system.

**Comment:** Synonym of heteropolar.



**Isopolar** (adj.) (Erdtman, 1947)

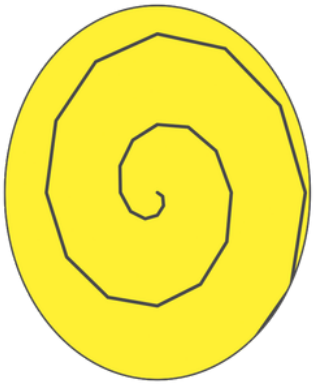
Describing a pollen grain or spore in which the proximal and distal faces of the exine are alike.



## S5. SYMMETRY

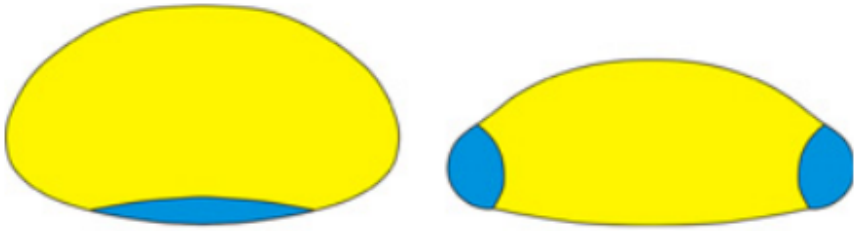
### Asymmetric

Describing pollen lacking any plane of symmetry.



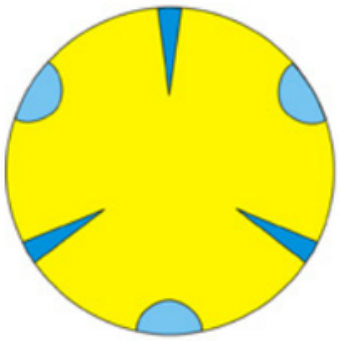
**Bilateral** (adj.)

Describing pollen and spores having a single, principal plane of symmetry.



**Radial** (adj.) (Nilsson and Muller, 1978)

Describing a pollen grain or spore with two or more vertical planes of symmetry, but, if only two such planes are present, then their axes are of equal length.



# SPORES

## A.1. Aperture type:

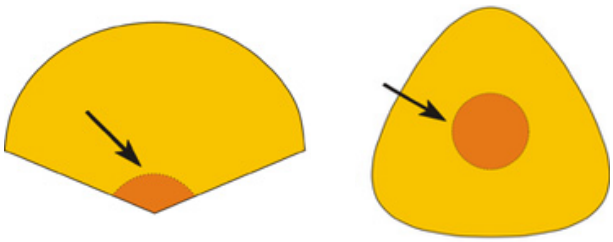
**Alete** (adj.) (Erdtman, 1943)

Describing a spore without a laesura (aperture).

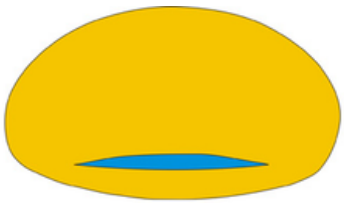
**Hilate** (adj.) (Erdtman, 1943)

Circular, indistinctly delimited, irregular aperture or thinning in spores

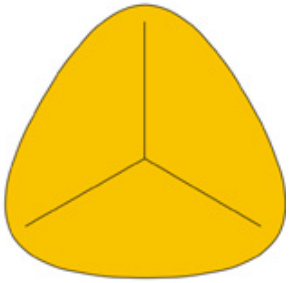
Comment: This kind of feature is present in certain **bryophytes** and fungi and may occur on the proximal or distal face.



**Monolete** (adj.) (Erdtman, 1943)
Describing a spore with a single laesura.



**Trilete** (adj.) (Erdtman, 1943)
Describing a spore with three l**aesurae**, thus showing a **trilete mark**.

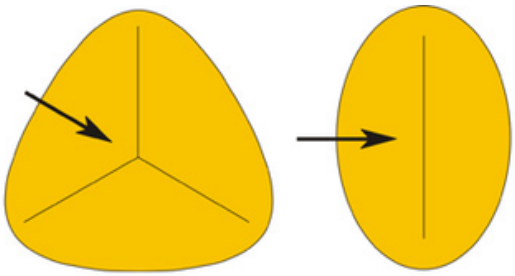


## A.2. Curvatura:

A line in spores extending from the extremities of the ends of the radii of the laesura and thus delimiting the contact areas.

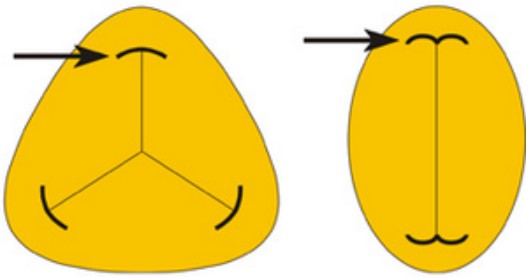
**Absent**

Absence of curvatura



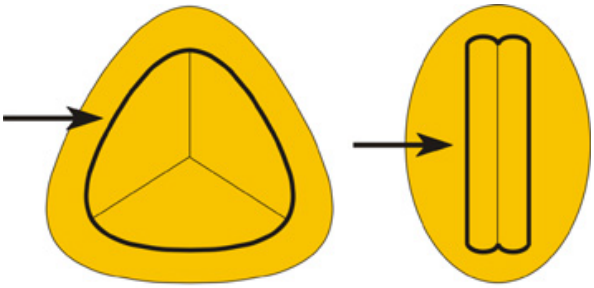
**Imperfect** (Potonié, 1934)

Present as forked extensions that do not join.



**Perfect** (Potonié, 1934)

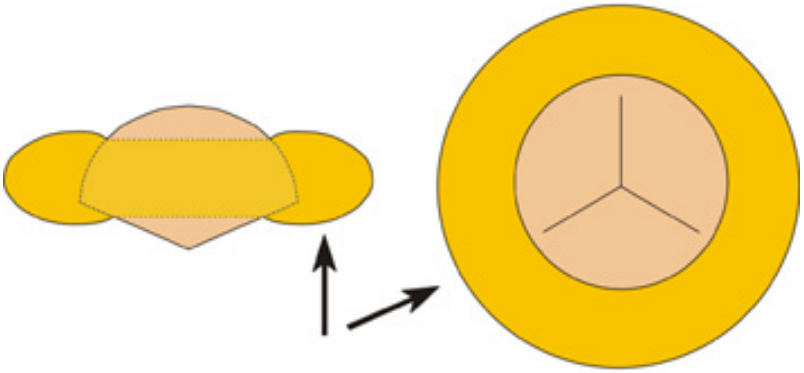
Continuous around the proximal face.



## A.3. Structural Modifications:

**Cingulate**(adj.) (Potonié & Kremp, 1955)

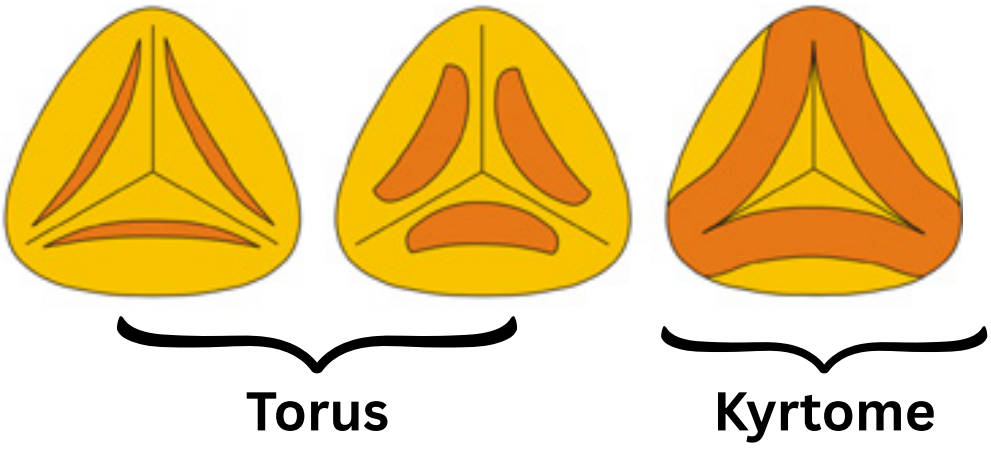
Describing a spore with a **cingulum**, a thick outer structure of a spore that projects at the equator, but does not extend over the distal or proximal face.



**Kyrtomate**(adj.) (Potonié & Kremp, 1955)

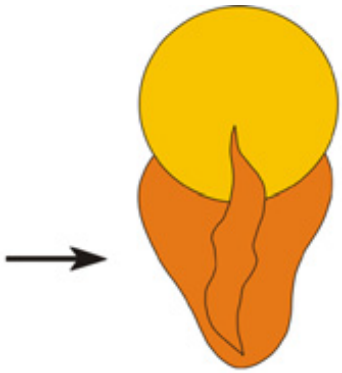
A spore withe a more or less arcuate fold or band in the **interradial** areas outside the laeusra of trilete spores.

Comment: Some palynologist prefer to use **torus** for separate interradial bands, and **kyrtome** for a connected feature.



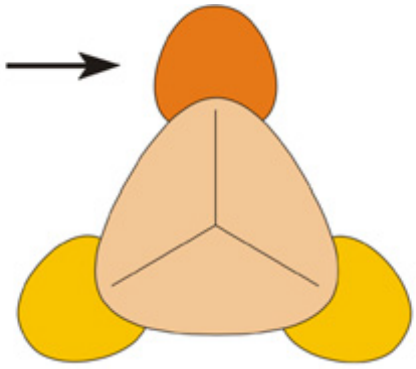
**Trifoliate**(adj.) (Potonié, 1956)

Describing a spore with a three-bladed, proximal feature arising from the proximal pole of a megaspore.



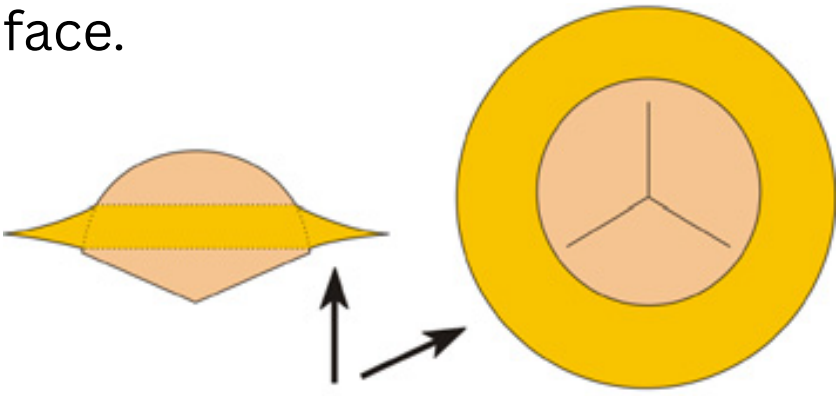
**Valvate**(adj.) (Potonié & Kremp, 1955)

Describing a spore with radial thickenings in the areas beyond the endes of the laesurae of trilete spores.



**Zonate**(adj.) (Potonié & Kremp, 1955)

A spore with a thin outer structure that projects at the equator, but does not extend over the distal face or proximal face.



## O.1. Exospore layers:

**1-layered**

When the exospore is composed by a single compact layer.

**2-layered**

When the exospore is composed by two layers (i.e. *inner exospore* & *outer exospore*).

Comment: Some spores can have more than 2 layers. However, in most cases that is only notorious in SEM microscopy.

## O.2. Perispore layers:

**Acavate** (adj.)

When the perispore layer is simple or compact, with absence of hollow cavities.

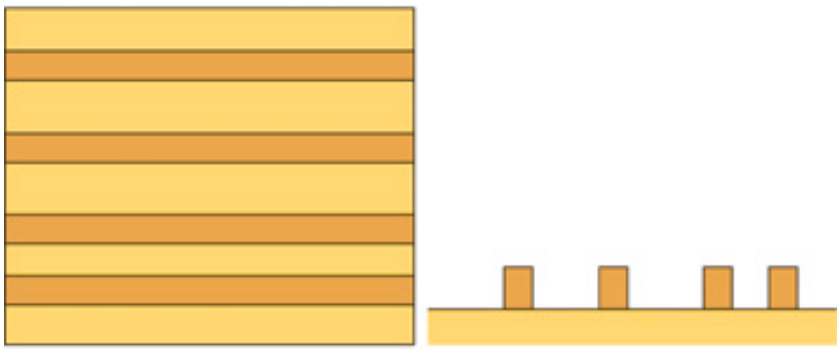
**Cavate** (adj.) (Tryon & Lugardon, 1991)

Perispore with hollow cavities separating inner and outer layers.

## O.3. Sculpture type:

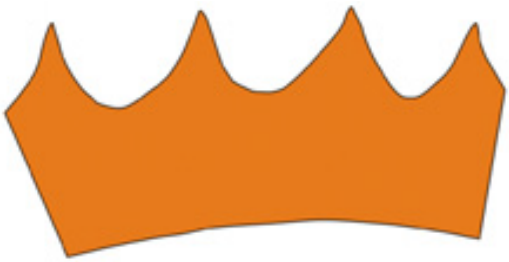
**Cicatricosate** (adj.) Potonié, 1934

Describing spores marked with scars.



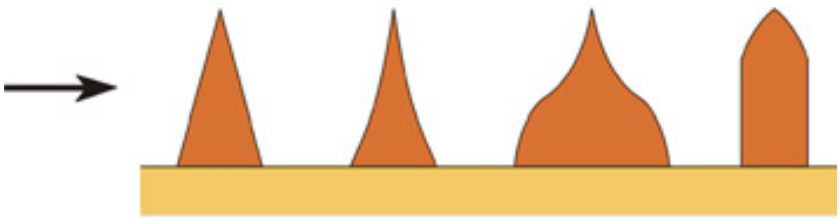
**Cristate** (adj.) (Potonié & Kremp, 1955)

A spore with crest-like ornamentation elements, taller that it is wide, characterized by a narrowly curved base and a sharp upper edge.



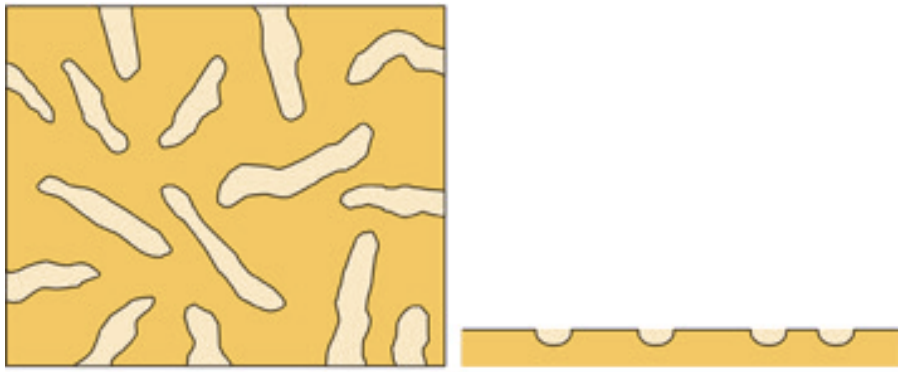
**Echinate** (adj.) (Wodehose, 1928)

Describing a spore with an ornamentation comprising spines longer than 1 µm.



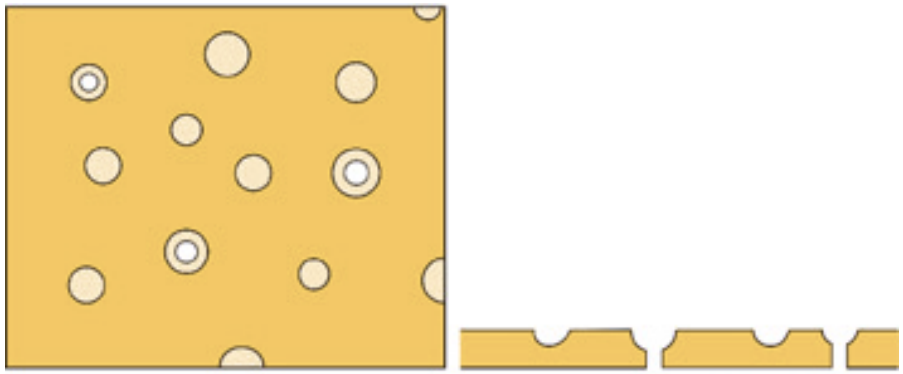
**Fossulate** (adj.) (Fægri and Iversen, 1950)

A spore featuring ornamentation consisting of an elongated, irregular groove in the surface.



**Foveolate** (adj.) (Erdtman, 1952)

A spore featuring ornamentation consisting of more or less rounded depressions or lumina more than 1 µm in diameter. The distance between foveolae is grater thant their breadth.



**Foveoreticulate** (adj.)

Describes a spore surface that has pits large enough and close enough together to form a network-like pattern.

**Gemmate** (adj.) (Iversen & Troels-Smith, 1950)

Describing a spore with sculpture elements thar are constricted at its base, higher than 1 µm, and that has approximately the same widht as its height.



**Granulate** (adj.) (Erdtman, 1952)

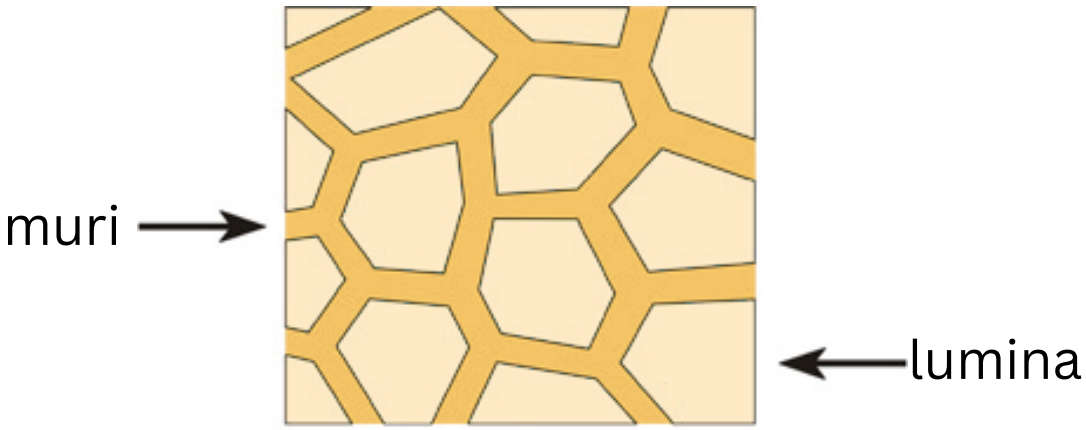
Spores with very small and rounded sculpture elements on its surface.

**Laevigate** (adj.) (Jackson, 1928)

Describing a spore with a smooth, plain exospore, as if polished.

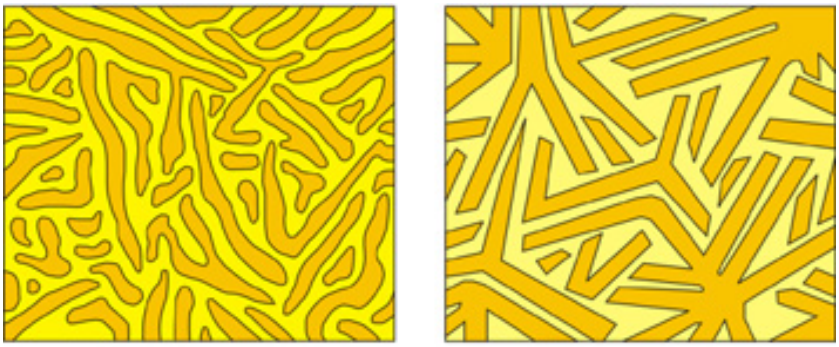
**Reticulate** (adj.) (Praglowski & Punt, 1973)

Describing a spore with a network-like pattern consisting of lumina or other spaces wider than 1 µm bordered by elements narrower than the lumina (**muri**).



**Rugulate** (adj.) (Iversen & Troels-Smith, 1950)

Describing a type of ornamentation consisting of elongated sculpture elements more than 1 µm long, arranged in an irregular pattern that is intermediate between striate and reticulate.

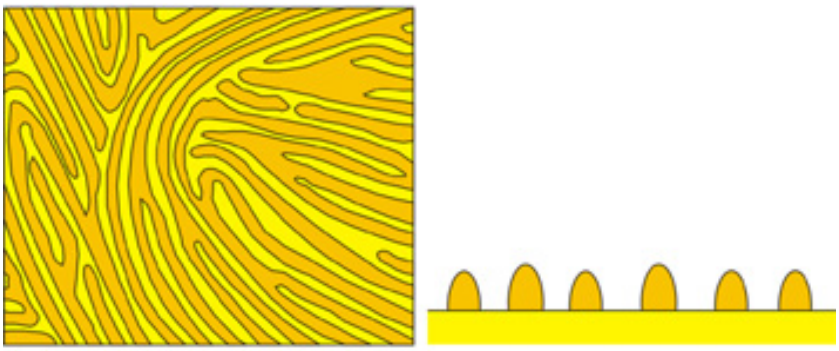


**Scabrate** (adj.) (Iversen & Troels-Smith, 1950)

Describing sculpture elements of any shape, smaller than 1 µm in all dirrections.

**Striate** (adj.) (Iversen & Troels-Smith, 1950)

Describing a condition with elongated, generally parallel elements separated by grooves.



**Verrucate** (adj.) (Iversen & Troels-Smith, 1950)

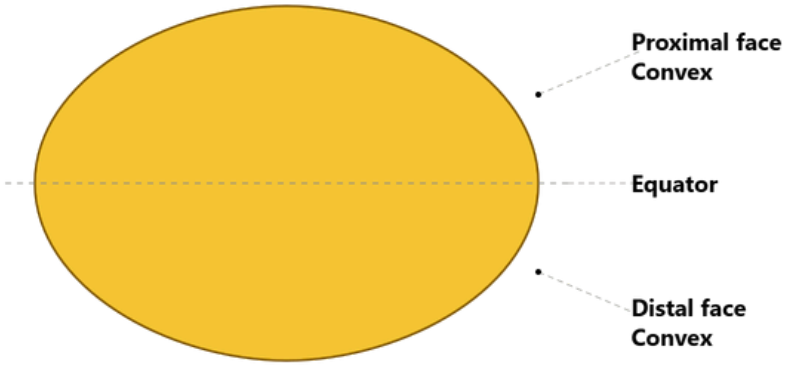
Describing a spore with wart-like sculpture elements, more than 1 µm wide, that is broader than it is high and is not constricted at the base.



# S.1. Lateral shape:

## Biconvex

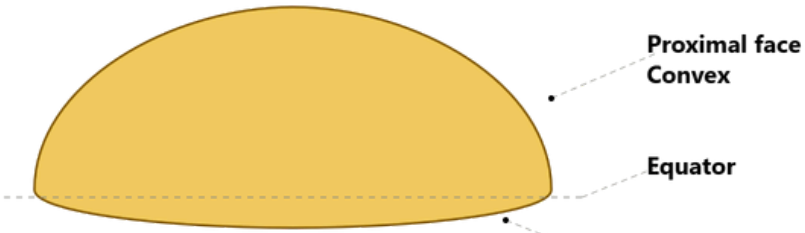
Both the proximal and distal faces curve outwards.




Both hemispheres bulge outward from the equatorial plane

## Plane-convex

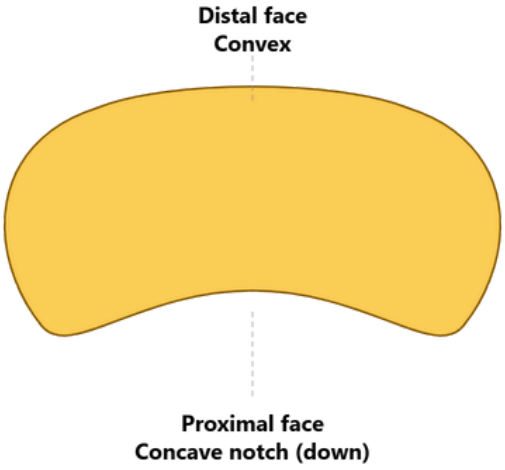
One of the poles curves out whilst the other one is flat.




Proximal side domes outward; distal side stays close to flat

## Reniform

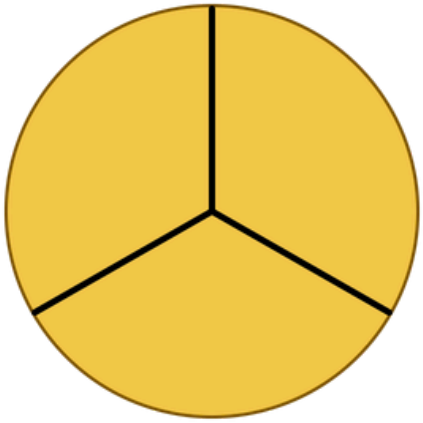
Bean-shaped spore, with a concave proximal face.



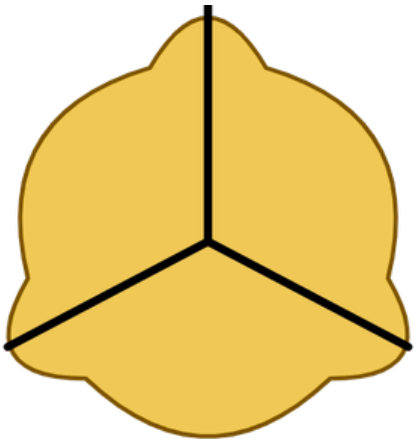

Bean-shaped outline: rounded convex top, indented concave bottom

## S.2. Lateral shape:

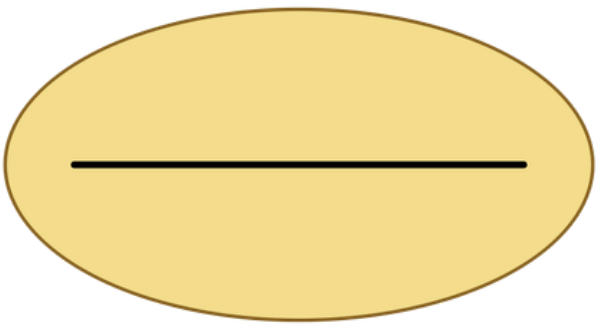
**Circular**
A spore with a round outline.



**Circular to Triangular-Obtuse-Convex**
Rounded outline, halfway between a circle and a triangle, with bulging sides.



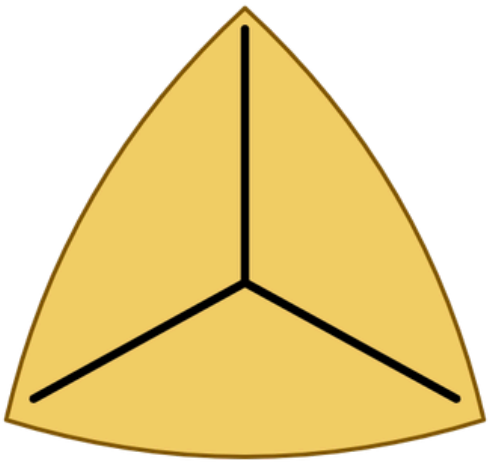
**Elliptic**
Oval outline, longer than it is wide.



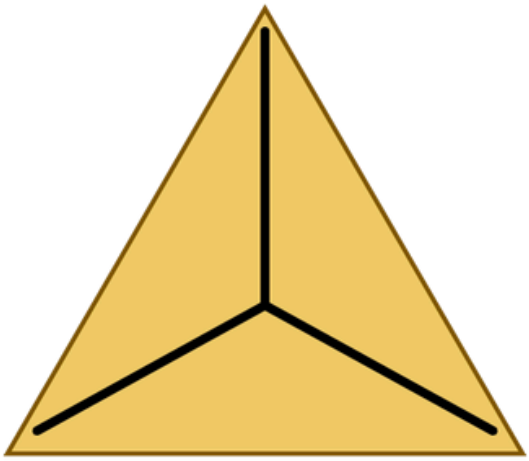
**Triangular-Acute-Concave**
Sharp angles, sides curved inwards.



**Triangular-Acute-Convex**
Sharp angles, sides bulged outwards.



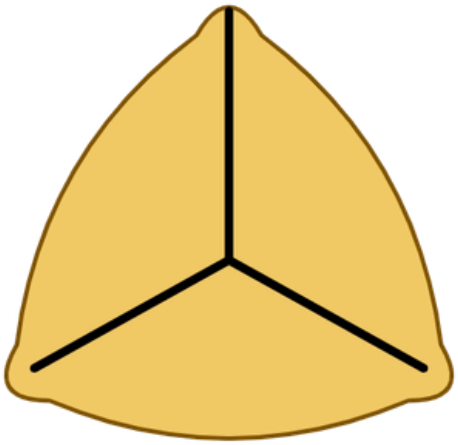
**Triangular-Acute-Straight**
Sharp angles, flat sides - a plain triangle.



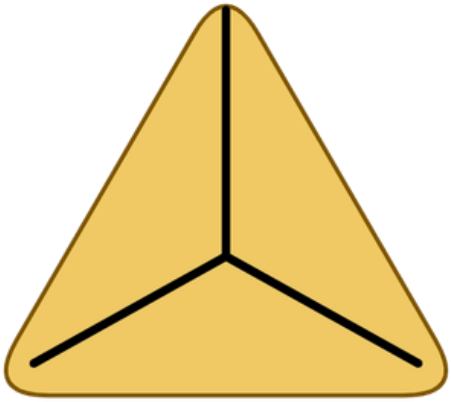
**Triangular-Obtuse-Concave**
Blunt rounded angles, sides curved inwards.



**Triangular-Obtuse-Convex**
Blunt rounded angles, sides bulged outwards.



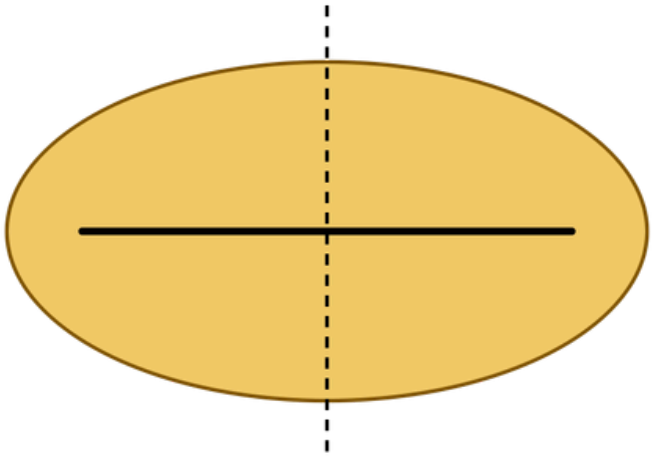
**Triangular-Obtuse-Straight**
Blunt rounded angles, flat sides.



## S.3. Symmetry:

**Bilateral** (adj.) (Walker & Doyle, 1975)

A spore that has a principal-main plane of symmetry.



**Radial** (adj.) (Nilsson & Muller, 1978)
Describing a spore with or two or more vertical planes of symmetry.

